

Advanced Curiosity Tasks (Style: "How Do Big Systems...")

- **Task 1: Database Structures Used by Big Companies**

Company	Database Type	Databases Used	Why This Structure?
Instagram	Relational (SQL) + NoSQL	PostgreSQL, Cassandra, Redis	PostgreSQL for user data, Cassandra for feed scalability, Redis for caching.
Uber	Relational + NoSQL	MySQL, Cassandra, Riak, Redis	MySQL for trips, Cassandra for real-time location, Redis for fast reads.
Amazon	NoSQL	DynamoDB (in-house), RDS for some SQL	DynamoDB handles massive scale and global availability, RDS for structured data.

Why Big Systems Use This Structure:

- SQL Databases (e.g., MySQL, PostgreSQL) are used for:
 - Structured, relational data (users, orders, transactions)
 - Strong consistency and integrity
- NoSQL Databases (e.g., Cassandra, DynamoDB, MongoDB) are used for:
 - High-speed read/write
 - Massive scalability (millions of users, real-time feeds)
 - Flexibility (unstructured data, denormalization)

Who Uses SQL Server?

Stack Overflow uses Microsoft SQL Server for its relational backend.

- **Task 2: How Do Online Games Store Player Data?**

Online games like PUBG, Fortnite, Call of Duty, and others store key player data such as:

- Player profiles
- Progress and level
- Match history
- Inventory and in-game purchases
- Leaderboard rankings

This data is stored on remote servers (game backends), often using a database system to manage and retrieve it quickly during gameplay.

What Kind of Databases Do Games Use?

- Relational (SQL) → User accounts, transactions, purchases, secure data → Structured, ACID-compliant, reliable.
- NoSQL (e.g. MongoDB, Redis, Cassandra) → Fast access to game state, leaderboards, session info → High-speed, scalable, flexible

Do They Use SQL Databases?

Yes, especially for persistent player data that requires structure and integrity.

Example of a Game Using a Relational Database:

- EVE Online (a large-scale space MMO):
 - Uses Microsoft SQL Server to store transactional and economic data.
 - Stores inventory, player actions, and financial market data in structured SQL tables.

• Task 3: How Are Databases Used in Offline Systems?

Yes — databases can work perfectly fine without the internet, and they are commonly used in offline systems such as military bases, airplanes, hospitals, and factories where connectivity may be limited or intentionally isolated for security.

Offline systems store data locally on the device or in an internal network:

- Military bases: Local servers store classified personnel data, mission plans, and logs — often in isolated (air-gapped) environments.
- Airplanes: Store flight data, diagnostics, passenger manifests, and offline entertainment metadata.

What Kind of Databases Are Used Offline?

SQL Server → Government, business, military apps
MySQL / PostgreSQL → Local servers in hospitals, ships
Oracle → Secure enterprise environments

Can SQL Server Work Without the Internet?

Yes, Microsoft SQL Server is a standalone system. It can be installed on a local machine or internal network without needing internet access. Once installed and licensed, it can manage, query, and store data fully offline.

- **Task 4: How Do E-commerce Platforms Organize Data?**

E-commerce sites use smart data organization techniques to speed up search, filtering, and recommendations. They use both SQL and Cloud.

Challenges in Storing Millions of Products

- **Scalability**

Need to handle millions of new products, daily price changes, inventory updates.

- **Fast Search**

Searching across millions of records with filters (brand, rating, price) requires search indexing.

- **High Traffic**

Must support millions of users at once — uses load balancing, distributed databases, and caching.

- **Real-time Sync**

Inventory and orders must sync across multiple systems in real time to prevent overselling.

- **Data Redundancy & Backup**

To avoid loss and ensure availability in different regions.

Why Are Online Stores So Fast?

Caching (Redis, CDN) → Stores frequently accessed data for instant load times.

Search Engines → Makes keyword and filter search ultra-fast.

Distributed Databases → Spreads load across servers to avoid bottlenecks.

Real Example: Amazon

Uses DynamoDB (NoSQL) for scale and performance

Uses RDS/Aurora (SQL) for order management

Uses Elasticsearch for product search

Relies heavily on AWS Cloud for storage, compute, and distribution

- **Task 5: How Do Banking Systems Protect Customer Data?**

Banks manage extremely sensitive data — like account numbers, balances, transactions, and personal details — so they rely on secure, robust database systems with multiple protection layers.

Key Ways Banks Protect Data Using Databases:

Encryption (at rest & in transit) → Data is encrypted using standards like AES-256 when stored and TLS when transmitted.

Access Control & Role Management → Only authorized users (like tellers, auditors) can access specific data.

Firewalls and VPNs → Secure internal networks protect database connections from outside attacks.

Examples of Banks Using Oracle or SQL Server:

- **HSBC → Oracle Database**

Task 6: "Draw Me the System!"

